

1. NAME OF THE MEDICINAL PRODUCT

Rotaqor Film Coated Tablet 10mg

Rotaqor Film Coated Tablet 20mg

Rotaqor Film Coated Tablet 40mg

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

Each 10 mg film-coated tablet contains 10 mg of atorvastatin (as crystalline calcium)

Each 20 mg film-coated tablet contains 20 mg of atorvastatin (as crystalline calcium)

Each 40 mg film-coated tablet contains 40 mg of atorvastatin (as crystalline calcium)

For a full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

Dosage form: Film-coated tablet.

10mg

White to off-white, round, biconvex film-coated tablet with debossment “A10” on one side, plain on the other side

20mg

White to off-white, round, biconvex film-coated tablet with debossment “A20” on one side and breaking notch on the other side

40mg

White to off-white, round, biconvex film-coated tablet with debossment “A40” on one side and cross scored breaking notch on the other side

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Atorvastatin is indicated as an adjunct to diet for the treatment of patients with elevated total cholesterol (total-C), low density lipoprotein cholesterol (LDL-C), apolipoprotein B (apo B), and triglycerides (TG) and to increase high density lipoprotein cholesterol (HDL-C) in patients with primary hypercholesterolemia (heterozygous familial and nonfamilial hypercholesterolemia), combined (mixed) hyperlipidemia (*Fredrickson* Types IIa and IIb), elevated serum TG levels (*Fredrickson* Type IV), and for patients with dysbetalipoproteinemia (*Fredrickson* Type III) who do not respond adequately to diet.

Atorvastatin is also indicated for the reduction of total-C and LDL-C in patients with homozygous familial hypercholesterolemia.

Prevention of Cardiovascular Disease

In adult patients without clinically evident cardiovascular disease (CVD), but with multiple risk factors for coronary heart disease (CHD) such as age, smoking, hypertension, low HDL-C, or a family history of early CHD, Rotaqor is indicated to:

- Reduce the risk of myocardial infarction (MI)
- Reduce the risk of stroke
- Reduce the risk for revascularization procedures and angina

In patients with type 2 diabetes, and without clinically evident CHD, but with multiple risk factors for coronary heart disease such as retinopathy, albuminuria, smoking, or hypertension, Rotaqor is indicated to:

- Reduce the risk of myocardial infarction
- Reduce the risk of stroke

In patients with clinically evident CHD, Rotaqor is indicated to:

- Reduce the risk of non-fatal MI

- Reduce the risk of fatal and non-fatal stroke,
- Reduce the risk for revascularization procedures,
- Reduce the risk of hospitalization for congestive heart failure (CHF)
- Reduce the risk of angina

Pediatric Patients (10-17 years of age)

Atorvastatin is indicated as an adjunct to diet to reduce total-C, LDL-C, and apo B levels in boys and postmenarchal girls, 10 to 17 years of age, with heterozygous familial hypercholesterolemia if after an adequate trial of diet therapy the following findings are present:

- a. LDL-C remains ≥ 190 mg/dL or
- b. LDL-C remains ≥ 160 mg/dL and:
 - There is a positive family history of premature CVD or
 - Two or more other CVD risk factors are present in the pediatric patient

4.2 Posology and method of administration

Posology

General

Before instituting therapy with atorvastatin, an attempt should be made to control hypercholesterolemia with appropriate diet, exercise and weight reduction in obese patients, and to treat the underlying medical problems. The patient should continue on a standard cholesterol-lowering diet during treatment with atorvastatin. The dosage range is 10 mg to 80 mg once daily. Doses may be given any time of the day, with or without food. Starting and maintenance dosage should be individualized according to baseline LDL-C levels, the goal of therapy, and patient response. After initiation and/or upon titration of atorvastatin, lipid levels should be analyzed within 2 to 4 weeks, and dosage adjusted accordingly.

Primary Hypercholesterolemia and Combined (Mixed) Hyperlipidemia

The majority of patients are controlled with 10 mg atorvastatin once daily. A therapeutic response is evident within 2 weeks, and the maximum response is usually achieved within 4 weeks. The response is maintained during chronic therapy.

Homozygous Familial Hypercholesterolemia

Most patients responded to 80 mg of atorvastatin with a greater than 15% reduction in LDL-C (18%-45%).

Heterozygous Familial Hypercholesterolemia in Pediatric Patients (10-17 years of age)

The recommended starting dose of atorvastatin is 10 mg/day; the usual dose range is 10 to 20 mg orally once daily. Doses should be individualized according to the recommended goal of therapy. Adjustments should be made at intervals of 4 weeks or more.

Severe Dyslipidemias in Pediatric Patients

For patients aged 10 years and above, the recommended starting dose is 10 mg atorvastatin once daily. The dose may be increased to 80 mg daily, according to the response and tolerability. Doses should be individualized according to the recommended goal of therapy. Adjustments should be made at intervals of 4 weeks or more.

Use in Patients with Hepatic Insufficiency

See section 4.3 (Contraindications) and section 4.4 (Special warnings and precautions for use).

Use in Patients with Renal Insufficiency

Renal disease has no influence on the plasma concentrations or on the LDL-C reduction with atorvastatin. Thus, no dose adjustment is required.

Use in the Elderly

No differences in safety, efficacy or lipid treatment goal attainment were observed between elderly patients and the overall population.

Dosage in Patients Taking Cyclosporine, Clarithromycin, Itraconazole, or Certain Protease Inhibitors

In patients taking cyclosporine or the HIV protease inhibitors (tipranavir plus ritonavir) or the hepatitis C protease inhibitor (telaprevir), therapy with atorvastatin should be avoided.

In patients with HIV taking lopinavir plus ritonavir, caution should be used when prescribing atorvastatin and the lowest dose necessary is employed.

In patients taking clarithromycin, itraconazole, or in patients with HIV taking a combination of saquinavir plus ritonavir, darunavir plus ritonavir, fosamprenavir, or fosamprenavir plus ritonavir, therapy with atorvastatin should be limited to 20 mg, and appropriate clinical assessment is recommended to ensure that the lowest dose necessary of atorvastatin is employed.

In patients taking the HIV protease inhibitor nelfinavir or the hepatitis C protease inhibitor boceprevir, therapy with atorvastatin should be limited to 40 mg, and appropriate clinical assessment is recommended to ensure that the lowest dose necessary of atorvastatin is employed.

Use in Children (Homozygous Familial Hypercholesterolemia)

Treatment experience in a pediatric population is limited to doses of atorvastatin up to 80mg/day for one year in 8 patients with homozygous FH. No clinical or biochemical abnormalities were reported in these patients.

Method of administration

Atorvastatin is for oral administration. Each daily dose of atorvastatin is given all at once and may be given at any time of day with or without food.

4.3 Contraindications

Atorvastatin is contraindicated in patients who have:

- Hypersensitivity to any component of this medication
- Active liver disease or unexplained persistent elevations of serum transaminases exceeding three times the upper limit of normal (ULN)

or who are:

- Pregnant, breast-feeding, or childbearing potential who are not using adequate contraceptive measures. Atorvastatin should be administered to women of childbearing age only when such patients are highly unlikely to conceive and have been informed of the potential hazards to the fetus.

4.4 Special warnings and precautions for use

Hepatic effects

As with other lipid-lowering agents of the same class, moderate ($>3 \times \text{ULN}$) elevations of serum transaminases have been reported following therapy with atorvastatin. Liver function was monitored during pre-marketing as well as post-marketing clinical studies of atorvastatin at doses of 10 mg, 20 mg, 40 mg and 80 mg.

Persistent increases in serum transaminases ($>3 \times \text{ULN}$ on two or more occasions) occurred in 0.7% of patients who received atorvastatin. The incidence of these abnormalities was 0.2%, 0.2%, 0.6%, and 2.3% for the 10 mg, 20 mg, 40 mg and 80 mg doses respectively. Increases were generally not associated with jaundice or other clinical signs or symptoms. When the dosage of atorvastatin was reduced, or drug treatment interrupted or discontinued, transaminase levels returned to pre-treatment levels. Most patients continued treatment on a reduced dose of atorvastatin without sequelae.

Liver function tests should be performed before the initiation of treatment and periodically thereafter. Patients who develop any signs or symptoms suggesting liver injury should have liver function tests performed. Patients who develop increased transaminase levels should be monitored until the abnormality(ies) resolve(s). Should an increase in ALT or AST of $>3 \times \text{ULN}$ persist, reduction of dose or withdrawal of atorvastatin is recommended. Atorvastatin can cause an elevation in transaminases.

Atorvastatin should be used with caution in patients who consume substantial quantities of alcohol and/or have a

history of liver disease. Active liver disease or unexplained persistent transaminase elevations are contraindications to the use of atorvastatin.

Skeletal Muscle Effects

Myalgia has been reported in atorvastatin-treated patients. Myopathy, defined as muscle ache or muscle weakness in conjunction with increases in creatine phosphokinase (CPK) values $>10 \times$ ULN, should be considered in any patient with diffuse myalgias, muscle tenderness or weakness, and/or marked elevation of CPK. Patients should be advised to promptly report unexplained muscle pain, tenderness or weakness, particularly if accompanied by malaise or fever. Atorvastatin therapy should be discontinued if markedly elevated CPK levels occur or if myopathy is diagnosed or suspected. The risk of myopathy is increased with concurrent administration of drugs that increase the systemic concentration of atorvastatin. Many of these drugs inhibit cytochrome P450 3A4 (CYP 3A4) metabolism and/or drug transport. CYP 3A4 is the primary hepatic isozyme known to be involved in the biotransformation of atorvastatin. Physicians considering combined therapy with atorvastatin and fibric acid derivatives, erythromycin, immunosuppressive drugs, azole antifungals, HIV/HCV protease inhibitors, HCV NS5A/NS5B inhibitors, letermovir or lipid-modifying doses of niacin ($\geq 1\text{g/day}$) should carefully weigh the potential benefits and risks and should carefully monitor patients for any signs and symptoms of muscle pain, tenderness, or weakness, particularly during the initial months of therapy and during any periods of upward dosage titration of either drug. Therefore, lower starting and maintenance doses of atorvastatin should also be considered when taken concomitantly with the aforementioned drugs. The concurrent use of atorvastatin and fusidic acid is not recommended, therefore, temporary suspension of atorvastatin is advised during fusidic acid therapy. Periodic CPK determinations may be considered in such situations, but there is no assurance that such monitoring will prevent the occurrence of severe myopathy. Atorvastatin may cause an elevation of CPK.

There have been very rare reports of an immune-mediated necrotizing myopathy (IMNM) during or after treatment with some statins. IMNM is clinically characterized by persistent proximal muscle weakness and elevated serum creatine kinase, which persist despite discontinuation of statin treatment, muscle biopsy showing necrotizing myopathy without significant inflammation; improvement with immunosuppressive agents.

As with other drugs in this class, rare cases of rhabdomyolysis with acute renal failure secondary to myoglobinuria, have been reported. A history of renal impairment may be a risk factor for the development of rhabdomyolysis. Such patients merit closer monitoring for skeletal muscle effects. Atorvastatin therapy should be temporarily withheld or discontinued in any patient with an acute, serious condition suggestive of a myopathy or with a risk factor predisposing to the development of renal failure secondary to rhabdomyolysis, (e.g., severe acute infection; hypotension; major surgery; trauma; severe metabolic, endocrine, and electrolyte disorders; and uncontrolled seizures).

Hemorrhagic Stroke

Patients without CHD who had a stroke or transient ischemic attack (TIA) within the preceding 6 months and were initiated on atorvastatin 80 mg revealed a higher incidence of hemorrhagic stroke. Patients with hemorrhagic stroke on entry appeared to be at increased risk for recurrent hemorrhagic stroke. However, in patients treated with atorvastatin 80 mg there were fewer strokes of any type and fewer CHD events.

Endocrine Function

Increases in hemoglobin A1c (HbA1c) and fasting serum glucose levels have been reported with 3 hydroxy-3-methylglutaryl-coenzyme A (HMG-CoA) reductase inhibitors, including atorvastatin. The risk of hyperglycemia, however, is outweighed by the reduction in vascular risk with statins.

Myasthenia Gravis/ Ocular Myasthenia

In few cases, statins have been reported to induce *de novo* or aggravate pre-existing myasthenia gravis or ocular myasthenia (see section 4.8). Atorvastatin should be discontinued in case of aggravation of symptoms. Recurrences when the same or a different statin was (re-) administered have been reported.

4.5 Interaction with other medicinal products and other forms of interaction

The risk of myopathy during treatment with HMG-CoA reductase inhibitors is increased with concurrent administration of cyclosporine, fibric acid derivatives, lipid-modifying doses of niacin or CYP3A4/transporter inhibitors (e.g., erythromycin and azole antifungals).

Inhibitors of CYP3A4

Atorvastatin is metabolized by CYP3A4. Concomitant administration of atorvastatin with inhibitors of CYP3A4 can lead to increases in plasma concentrations of atorvastatin. The extent of interaction and potentiation of effects depend on the variability of effect on CYP3A4.

Erythromycin/Clarithromycin: Co-administration of atorvastatin with erythromycin (500 mg four times daily) or clarithromycin (500 mg twice daily), known inhibitors of CYP3A4, was associated with higher plasma concentrations of atorvastatin.

Protease Inhibitors: Co-administration of atorvastatin with protease inhibitors, known inhibitors of CYP3A4, was associated with increased plasma concentrations of atorvastatin.

Diltiazem Hydrochloride: Co-administration of atorvastatin (40 mg) with diltiazem (240 mg) was associated with higher plasma concentrations of atorvastatin.

Cimetidine: No clinically significant interactions were seen.

Itraconazole: Concomitant administration of atorvastatin (20-40 mg) with itraconazole (200 mg) was associated with an increase in atorvastatin AUC.

Grapefruit Juice: Contains one or more components that inhibit CYP 3A4 and can increase plasma concentrations of atorvastatin, especially with excessive grapefruit juice consumption (>1.2 L/day).

Transporter Inhibitors: Atorvastatin is a substrate of the hepatic transporters. Concomitant administration of atorvastatin 10 mg and cyclosporine 5.2 mg/kg/day resulted in an increase in exposure to atorvastatin. Cyclosporine is an inhibitor of organic anion-transporting polypeptide 1B1 (OATP1B1), OATP1B3, multi-drug resistance protein 1 (MDR1), and breast cancer resistance protein (BCRP) as well as CYP3A4, thus it increases exposure to atorvastatin.

Glecaprevir and pibrentasvir are inhibitors of OATP1B1, OATP1B3, MDR1 and BCRP, thus they increase exposure to atorvastatin. Do not exceed 10 mg atorvastatin daily.

Concomitant administration of atorvastatin 20 mg and letermovir 480 mg daily resulted in an increase in exposure to atorvastatin. Letermovir inhibits efflux transporters P-gp, BCRP, MRP2, OAT2 and hepatic transporter OATP1B1/1B3, thus it increases exposure to atorvastatin. Do not exceed 20 mg atorvastatin daily.

The magnitude of CYP3A- and OATP1B1/1B3-mediated drug interactions on co-administered drugs may be different when letermovir is co-administered with cyclosporine. Use of atorvastatin should be avoided in patients taking letermovir co-administered with cyclosporine.

Elbasvir and grazoprevir are inhibitors of OATP1B1, OATP1B3, MDR1 and BCRP, thus they increase exposure to atorvastatin. Use with caution and lowest dose necessary. Dose of atorvastatin should not exceed 20 mg daily in patients receiving concomitant medications with products containing elbasvir and grazoprevir.

Inducers of CYP3A4: Concomitant administration of atorvastatin with inducers of CYP 3A4 (e.g., efavirenz, rifampin) can lead to variable reductions in plasma concentrations of atorvastatin. Due to the dual interaction mechanism of rifampin (CYP 3A4 induction and inhibition of hepatocyte uptake transporter OATP1B1), simultaneous co-administration of atorvastatin with rifampin is recommended, as delayed administration of atorvastatin after administration of rifampin has been associated with a significant reduction in atorvastatin plasma concentrations.

Antacids: Co-administration of atorvastatin with an oral antacid suspension containing magnesium and aluminum hydroxides decreased atorvastatin plasma concentrations; however, LDL-C reduction was not altered.

Antipyrine: Because atorvastatin does not affect the pharmacokinetics of antipyrine, interactions with other drugs metabolized via the same cytochrome isozymes are not expected.

Colestipol: Plasma concentrations of atorvastatin were lower (ratio of concentration: 0.74) when colestipol was administered with atorvastatin. However, lipid effects were greater when atorvastatin and colestipol were co-

administered than when either drug was given alone.

Digoxin: When multiple doses of digoxin and 10 mg atorvastatin were co-administered, steady-state plasma digoxin concentrations were unaffected. However, digoxin concentrations increased following administration of digoxin with 80 mg atorvastatin daily. Patients taking digoxin should be monitored appropriately.

Azithromycin: Co-administration of atorvastatin (10 mg once daily) with azithromycin (500 mg once daily) did not alter the plasma concentrations of atorvastatin.

Oral Contraceptives: Co-administration of atorvastatin with an oral contraceptive containing norethindrone and ethinyl estradiol increased the area under the concentration vs. time curve (AUC) values for norethindrone and ethinyl estradiol. These increases should be considered when selecting an oral contraceptive for a woman taking atorvastatin.

Warfarin: An atorvastatin interaction study with warfarin was conducted, and no clinically significant interactions were seen.

Colchicine: Although interaction studies with atorvastatin and colchicine have not been conducted, cases of myopathy have been reported with atorvastatin co-administered with colchicine, and caution should be exercised when prescribing atorvastatin with colchicine.

Amlodipine: In a drug-drug interaction study in healthy subjects, co-administration of atorvastatin 80 mg with amlodipine 10 mg resulted in an increase in exposure to atorvastatin which was not clinically meaningful.

Fusidic Acid: Although interaction studies with atorvastatin and fusidic acid have not been conducted, there is an increased risk of rhabdomyolysis in patients receiving a combination of statins, including atorvastatin, and fusidic acid. The mechanism of this interaction is not known. In patients where the use of systemic fusidic acid is considered essential, statin treatment should be discontinued throughout the duration of fusidic acid treatment. Statin therapy may be re-introduced seven days after the last dose of fusidic acid.

In exceptional circumstances, where prolonged systemic fusidic acid is needed, e.g., for the treatment of severe infections, the need for co-administration of atorvastatin and fusidic acid should only be considered on a case by case basis and under close medical supervision. The patient should be advised to seek medical advice immediately if they experience any symptoms of muscle weakness, pain or tenderness.

Fibrates: Concurrent use of fibrates may cause severe myositis and myoglobinuria.

Other Concomitant Therapy: Atorvastatin was used concomitantly with antihypertensive agents and estrogen replacement therapy without evidence of clinically significant adverse interactions. Interaction studies with specific agents have not been conducted.

4.6 Pregnancy and lactation

Atorvastatin is contraindicated in pregnancy. Women of childbearing potential should use adequate contraceptive measures. Atorvastatin should be administered to women of childbearing age only when such patients are highly unlikely to conceive and have been informed of the potential hazards to the fetus.

Atorvastatin is contraindicated while breast-feeding. It is not known whether this drug is excreted in human milk. Because of the potential for adverse reactions in nursing infants, women taking atorvastatin should not breast-feed.

4.7 Effects on ability to drive and use machines

None known.

4.8 Undesirable effects

Atorvastatin is generally well tolerated. Adverse reactions have usually been mild and transient.

The most frequent ($\geq 1\%$) adverse effects that may be associated with atorvastatin therapy, reported in patients participating in placebo-controlled clinical studies include:

Infections and infestations: nasopharyngitis

Metabolism and nutrition disorders: hyperglycaemia

Respiratory, thoracic and mediastinal disorders: pharyngolaryngeal pain, epistaxis

Gastrointestinal disorders: diarrhoea, dyspepsia, nausea, flatulence

Musculoskeletal and connective tissue disorders: arthralgia, pain in extremity, musculoskeletal pain, muscle spasms, myalgia, joint swelling

Investigations: liver function test abnormal, blood creatine phosphokinase increased

Additional adverse effects reported in atorvastatin placebo-controlled clinical trials include:

Psychiatric disorders: nightmare

Eye disorders: vision blurred

Ear and labyrinth disorders: tinnitus

Gastrointestinal disorders: abdominal discomfort, eructation

Hepatobiliary disorders: hepatitis, cholestasis

Skin and subcutaneous tissue disorders: urticaria

Musculoskeletal and connective tissue disorders: muscle fatigue, neck pain

General disorders and administration site conditions: malaise, pyrexia

Investigations: white blood cells urine positive

Frequency not known: Immune-mediated necrotizing myopathy

Not all effects listed above have been causally associated with atorvastatin therapy.

Pediatric Patients

Patients treated with atorvastatin had an adverse experience profile generally similar to that of patients treated with placebo, the most common adverse experiences observed in both groups, regardless of causality assessment, were infections.

No clinically significant effect on growth and sexual maturation was observed in a 3-year study in children ages 6 and above based on the assessment of overall maturation and development, assessment of Tanner Stage, and measurement of height and weight. The safety and tolerability profile in pediatric patients was similar to the known safety profile of atorvastatin in adult patients.

Post-marketing Experience

In post-marketing experience, the following additional undesirable effects have been reported:

Blood and lymphatic system disorders: thrombocytopenia;

Immune system disorders: allergic reactions (including anaphylaxis);

Injury, poisoning, and procedural complications: tendon rupture;

Metabolism and nutrition disorders: weight gain;

Nervous system disorders: hypoaesthesia, amnesia, dizziness, dysgeusia, myasthenia gravis;

Gastrointestinal disorders: pancreatitis;

Skin and subcutaneous tissue disorders: Stevens-Johnson syndrome, toxic epidermal necrolysis, angioedema, erythema multiforme, bullous rashes;

Musculoskeletal and connective tissue disorders: rhabdomyolysis, immune mediated necrotizing myopathy, myositis, back pain;

General disorders and administration site conditions: chest pain, peripheral oedema, fatigue.

Eye disorders: ocular myasthenia.

There have been rare post-marketing reports of cognitive impairment (e.g., memory loss, forgetfulness, amnesia, memory impairment, confusion) associated with statin use. These cognitive issues have been reported for all statins. The reports are generally non-serious and reversible upon statin discontinuation, with variable times to symptom onset (1 day to years) and symptom resolution (median 3 weeks). Increases in HbA1c and fasting blood glucose have been reported with statins. The risk of hyperglycemia, however, is outweighed by the reduction in vascular risk with statins.

4.9 Overdose

There is no specific treatment for atorvastatin overdose. Should an overdose occur, the patient should be treated symptomatically and supportive measures instituted, as required. Due to extensive drug binding to plasma proteins, hemodialysis is not expected to significantly enhance atorvastatin clearance.

5. PHARMACOLOGICAL PROPERTIES

5.1 Pharmacodynamic properties

Atorvastatin calcium is a synthetic lipid-lowering agent, which is an inhibitor of HMG-CoA reductase. This enzyme catalyzes the conversion of HMG-CoA to mevalonate, an early and rate-limiting step in cholesterol biosynthesis.

The empirical formula of atorvastatin calcium is $(C_{33}H_{34}FN_2O_5)_2Ca \cdot 3H_2O$ and its molecular weight is 1209.42.

Atorvastatin calcium is a white to off-white crystalline powder, practically insoluble in aqueous solutions of pH 4 and below. It is very slightly soluble in distilled water, pH 7.4 phosphate buffer, and acetonitrile; slightly soluble in ethanol; and freely soluble in methanol.

Mechanism of Action

Atorvastatin is a selective, competitive inhibitor of HMG-CoA reductase, the rate-limiting enzyme that converts **HMG-CoA** to mevalonate, a precursor of sterols, including cholesterol. In patients with homozygous and heterozygous familial hypercholesterolemia, nonfamilial forms of hypercholesterolemia, and mixed dyslipidemia, atorvastatin reduces total-C, LDL-C, and apo B. Atorvastatin also reduces very low-density lipoprotein cholesterol (VLDL-C) and TG and produces variable increases in HDL-C.

Atorvastatin lowers plasma cholesterol and lipoprotein levels by inhibiting HMG-CoA reductase and cholesterol synthesis in the liver and by increasing the number of hepatic LDL receptors on the cell surface for enhanced uptake and catabolism of LDL.

Atorvastatin reduces LDL production and the number of LDL particles. Atorvastatin produces a profound and sustained increase in LDL receptor activity coupled with a beneficial change in the quality of circulating LDL particles. Atorvastatin is effective in reducing LDL in patients with homozygous familial hypercholesterolemia, a population that has not normally responded to lipid-lowering medication.

Atorvastatin and some of its metabolites are pharmacologically active in humans. The primary site of action of atorvastatin is the liver, which is the principal site of cholesterol synthesis and LDL clearance. LDL-C reduction correlates better with drug dose than it does with systemic drug concentration. Individualization of drug dosage should be based on therapeutic response.

5.2 Pharmacokinetic properties

Absorption

Atorvastatin is rapidly absorbed after oral administration; maximum plasma concentrations occur within 1 to 2 hours. The extent of absorption and plasma atorvastatin concentrations increases in proportion to atorvastatin dose. Atorvastatin tablets are 95% to 99% bioavailable compared to solutions. The absolute bioavailability of atorvastatin is approximately 14% and the systemic availability of HMG-CoA reductase inhibitory activity is approximately 30%. The low systemic availability is attributed to presystemic clearance in gastrointestinal mucosa and/or hepatic first-pass metabolism. Although food decreases the rate and extent of drug absorption by approximately 25% and 9%, respectively, as assessed by C_{max} and AUC, LDL-C reduction is similar whether atorvastatin is given with or without food. Plasma atorvastatin concentrations are lower (approximately 30% for C_{max} and AUC) following evening drug administration compared to morning. However, LDL-C reduction is the same regardless of the time of day of drug administration.

Distribution

Mean volume of distribution of atorvastatin is approximately 381 liters. Atorvastatin is $\geq 98\%$ bound to plasma proteins. A red blood cell/plasma ratio of approximately 0.25 indicates poor drug penetration into red blood cells.

Metabolism

Atorvastatin is extensively metabolized to ortho- and parahydroxylated derivatives and various beta-oxidation products. *In vitro* inhibition of HMG-CoA reductase by ortho- and parahydroxylated metabolites is equivalent to that of atorvastatin. Approximately 70% of circulating inhibitory activity for HMG-CoA reductase is attributed to active metabolites. *In vitro* studies suggest the importance of atorvastatin metabolism by hepatic CYP3A4, consistent with increased plasma concentrations of atorvastatin in humans following co-administration with erythromycin, a known inhibitor of this isozyme. *In vitro* studies also indicate that atorvastatin is a weak inhibitor of CYP3A4. Atorvastatin coadministration did not produce a clinically significant effect in plasma concentrations of terfenadine, a compound predominantly metabolized by CYP3A4; therefore, it is unlikely that atorvastatin will significantly alter the pharmacokinetics of other CYP3A4 substrates. In animals, the ortho-hydroxy metabolite undergoes further glucuronidation.

Excretion

Atorvastatin and its metabolites are eliminated primarily in bile following hepatic and/or extrahepatic metabolism; however, the drug does not appear to undergo enterohepatic recirculation. Mean plasma elimination half-life of atorvastatin in humans is approximately 14 hours, but the half-life of inhibitory activity for HMG-CoA reductase is 20 to 30 hours due to the contribution of active metabolites. Less than 2% of a dose of atorvastatin is recovered in urine following oral administration.

Atorvastatin is a substrate of the hepatic transporters, OATP1B1 and OATP1B3 transporter. Metabolites of atorvastatin are substrates of OATP1B1. Atorvastatin is also identified as a substrate of the efflux transporters MDR1 and BCRP, which may limit the intestinal absorption and biliary clearance of atorvastatin.

Special Populations

Elderly

Plasma concentrations of atorvastatin are higher (approximately 40% for C_{max} and 30% for AUC) in healthy, elderly subjects (aged ≥ 65 years) than in young adults. No differences in safety, efficacy or lipid treatment goal attainment were observed between elderly patients and the overall population.

Pediatric population

Apparent oral clearance of atorvastatin in pediatric subjects appeared similar to adults when scaled allometrically by body weight. Consistent decreases in LDL-C and TC were observed over the range of atorvastatin and o-hydroxyatorvastatin exposures.

Gender

Plasma concentrations of atorvastatin in women differ (approximately 20% higher for C_{max} and 10% lower for AUC). From those in men, However, there were no clinical significance, resulting in no clinically significant differences in lipid effects among men and women.

Renal Insufficiency

Renal disease has no influence on the plasma concentrations or lipid effects of atorvastatin. Thus, dose adjustment in patients with renal dysfunction is not necessary.

Hemodialysis

While studies have not been conducted in patients with end-stage renal disease, hemodialysis is not expected to significantly enhance clearance of atorvastatin since the drug is extensively bound to plasma proteins.

Hepatic Insufficiency

Plasma concentrations of atorvastatin are markedly increased (approximately 16-fold in C_{max} and 11-fold in AUC) in patients with chronic alcoholic liver disease (Child-Pugh Class B).

Drug Interactions

The effect of co-administered drugs on the pharmacokinetics of atorvastatin as well as the effect of atorvastatin on the pharmacokinetics of co-administered drugs are summarized below (see section 4.4 Special warnings and precautions for use and section 4.5 Interaction with other medicinal products and other forms of interaction).

Effect of Co-administered Drugs on the Pharmacokinetics of Atorvastatin

Co-administered Drug and Dosing Regimen	Atorvastatin		
	Dose (mg)	Ratio of AUC^{&}	Ratio of C_{max}^{&}
[#] Cyclosporine 5.2 mg/kg/day, stable dose	10 mg QD ^a for 28 days	8.7	10.7
[#] Tipranavir 500 mg BID ^b /ritonavir 200 mg BID ^b , 7 days	10 mg SD ^c	9.4	8.6
[#] Glecaprevir 400 mg QD ^a /Pibrentasvir 120 mg QD ^a , 7 days	10 mg QD ^a for 7 days	8.3	22.0
[#] Telaprevir 750 mg q8h ^f , 10 days	20 mg SD ^c	7.9	10.6
[#] Elbasvir 50 mg QD ^a /grazoprevir 200 mg QD ^a , 13 days	10 mg SD ^c	1.95	4.3
[#] Boceprevir 800 mg TID ^d , 7 days	40 mg SD ^c	2.3	2.7
[#] Simeprevir 150 mg QD ^a , 10 days	40 mg SD ^c	2.12	1.7
[#] Lopinavir 400 mg BID ^b /ritonavir 100 mg BID ^b , 14 days	20 mg QD ^a for 4 days	5.9	4.7
^{#,*} Saquinavir 400 mg BID ^b / ritonavir 400 mg BID ^b , 15 days	40 mg QD ^a for 4 days	3.9	4.3
[#] Clarithromycin 500 mg BID ^b , 9 days	80 mg QD ^a for 8 days	4.5	5.4

#Darunavir 300 mg BID ^b / ritonavir 100 mg BID ^b , 9 days	10 mg QD ^a for 4 days	3.4	2.2
#Itraconazole 200 mg QD ^a , 4 days	40 mg SD ^c	3.3	1.20
#Letermovir 480 mg QD, 10 days ^a	20 mg SD ^c	3.29	2.17
#Fosamprenavir 700 mg BID ^b / ritonavir 100 mg BID ^b , 14 days	10 mg QD ^a for 4 days	2.5	2.8
#Fosamprenavir 1400 mg BID ^b , 14 days	10 mg QD ^a for 4 days	2.3	4.0
#Nelfinavir 1250 mg BID ^b , 14 days	10 mg QD ^a for 28 days	1.74	2.2
#Grapefruit juice, 240 mL QD ^{a*}	40 mg SD ^c	1.37	1.16
Diltiazem 240 mg QD ^a , 28 days	40 mg SD ^c	1.51	1.00
Erythromycin 500 mg QID ^c , 7 days	10 mg SD ^c	1.33	1.38
Amlodipine 10 mg, single dose	80 mg SD ^c	1.18	0.91
Cimetidine 300 mg QID ^c , 2 weeks	10 mg QD ^a for 2 weeks	1.00	0.89
Colestipol 10 g BID ^b , 24 weeks	40 mg QD ^a for 8 weeks	NA	0.74**
Maalox TC [®] 30 mL QID ^c , 17 days	10 mg QD ^a for 15 days	0.66	0.67
Efavirenz 600 mg QD ^a , 14 days	10 mg for 3 days	0.59	1.01
#Rifampin 600 mg QD ^a , 7 days (co-administered) [†]	40 mg SD ^c	1.12	2.9
#Rifampin 600 mg QD ^a , 5 days (doses separated) [†]	40 mg SD ^c	0.20	0.60
#Gemfibrozil 600 mg BID ^b , 7 days	40 mg SD ^c	1.35	1.00
#Fenofibrate 160 mg QD ^a , 7 days	40 mg SD ^c	1.03	1.02

& Represents ratio treatments (co-administered drug plus atorvastatin vs. atorvastatin alone).

See section **4.4 Special warnings and precautions for use** and section **4.5 Interaction with other medicinal products and other forms of interaction** for clinical significance.

* Greater increases in AUC (ratio of AUC up to 2.5) and/or C_{max} (ratio of C_{max} up to 1.71) have been reported with excessive grapefruit consumption (≥750 mL - 1.2 L/day).

** Ratio based on a single sample taken 8-16 h post dose.

† Due to the dual interaction mechanism of rifampin, simultaneous co-administration of atorvastatin with rifampin is recommended, as delayed administration of atorvastatin after administration of rifampin has been associated with a significant reduction in atorvastatin plasma concentrations.

‡ The dose of saquinavir / ritonavir in this study is not the clinically used dose. The increase in atorvastatin exposure when used clinically is likely to be higher than what was observed in this study. Therefore caution should be exercised and the lowest dose necessary should be used.

^a Once daily

^b Twice daily

^c Single dose

^d Three times daily

^e Four times daily

^f Every 8 hours

Effect of Atorvastatin on the Pharmacokinetics of Co-administered Drugs

Atorvastatin	Co-administered Drug and Dosing Regimen		
	Drug/Dose (mg)	Ratio of AUC ^{&}	Ratio of C _{max} ^{&}
80 mg QD ^a for 15 days	Antipyrine 600 mg SD ^c	1.03	0.89
80 mg QD ^a for 10 days	Digoxin 0.25 mg QD ^a , 20 days [#]	1.15	1.20
40 mg QD ^a for 22 days	Oral contraceptive QD ^a , 2 months		
	- Norethindrone 1 mg - Ethinyl estradiol 35 µg	1.28 1.19	1.23 1.30
10 mg SD ^c	Tipranavir 500 mg BID ^b /ritonavir 200 mg BID ^b , 7 days	1.08	0.96
10 mg QD ^a for 4 days	Fosamprenavir 1400 mg BID ^b , 14 days	0.73	0.82
10 mg QD ^a for 4 days	Fosamprenavir 700 mg BID ^b /ritonavir 100 mg BID ^b , 14 days	0.99	0.94

[&]Represents ratio treatments (co-administered drug plus atorvastatin vs. atorvastatin alone)

[#] See section **4.5 Interaction with other medicinal products and other forms of interaction** for clinical significance.

^a Once daily

^b Twice daily

^c Single dose

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Core

Cellulose microcrystalline

Calcium carbonate

Lactose monohydrate

Croscarmellose sodium

Hydroxypropylcellulose

Polysorbate 80

Magnesium stearate

Coating

Hypromellose 2910

Polyethylene glycol 6000

Titanium dioxide

- This medicinal product contains less than 1mmol (23mg) sodium per capsule, that is to say essentially “sodium-free”.
- If you have been told by your doctor that you have an intolerance to some sugars, contact your doctor before taking this medicinal product.

6.2 Shelf-life

2 years.

6.3 Special precautions for storage

Store below 30°C. Protect from light and moisture.

6.4 Nature and contents of container

Rotaqor film-coated tablets are packed in OPA/AL/PVC/Aluminium blisters (Al/Al blisters). The blisters are packed into carton boxes along with the package insert.

Pack size of 30's, 60's and 100's film-coated tablets. Not all pack sizes may be marketed.

6.5 Special precautions for disposal

Any unused medicinal product or waste material should be disposed of in accordance with local requirements.

7. PRODUCT REGISTRATION HOLDER

Sandoz Products Malaysia Sdn. Bhd.
Unit 1202, Level 12, Uptown 1,
No. 1, Jalan SS 21/58, Damansara Uptown,
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8. DATE OF REVISION OF THE TEXT

May 2025